

1. All questions are **Compulsory**.
2. All questions carry **equal** marks.
3. Draw **neat and labeled diagram** wherever necessary

Q.1. Attempt the following (Any Three)**[15MARKS]**

- a) Explain structure of C program.
- b) What is Compiler, Linker and preprocessor in C?
- c) What are the basic data types supported in C Programming Language?
- d) Explain Variable and Constant with example
- e) Explain typedef and typecasting in C.
- f) Define flowchart with its symbols.

Q.2. Attempt the following (Any Three)**[15MARKS]**

- a) What are the different types of operators in C? Explain.
- b) What are the various looping statement in C? Explain.
- c) Write note on break and continue statement.
- d) Explain Arithmetic operator with example.
- e) Write a C program to find whether the input number is even or odd.
- f) Explain Goto and Labels in C.

Q.3. Attempt the following (Any Three)**[15MARKS]**

- a) What is function? How function is defined?
- b) What is recursion? Explain with suitable example?
- c) Write a program to find reverse of a number.
- d) Explain different input and output function.
- e) Explain error handling. What is StdErr?
- f) Write note on header files.

Q.4. Attempt the following (Any Three)**[15MARKS]**

- a) What is pointer explain uses of pointer in C?
- b) What are * and & operators means?
- c) Write short note on dynamic memory allocation.
- d) Explain pointer to an array with example.
- e) What is command line argument in C?
- f) Write a C program to add two float numbers using pointers.

Q.5. Attempt the following (Any Three)**[15MARKS]**

- a) Define structure in C.
- b) Differentiate between structure and unions.
- c) How errors are handle in C.
- d) Define opening and closing files.
- e) Write note on pointer to structure in C.
- f) How to create an array of a structure? Explain with example.
